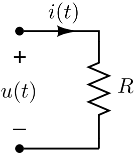
$$u(t) = \sqrt{2}U\cos(\omega t + \varphi_u) = \Re e[\sqrt{2}e^{j\omega t}\mathcal{U}]$$
$$i(t) = \sqrt{2}I\cos(\omega t + \varphi_i) = \Re e[\sqrt{2}e^{j\omega t}\mathcal{I}]$$




$$\boxed{u(t) = R \cdot i(t)}$$

$$\Downarrow$$

$$\Re e[\sqrt{2}e^{j\omega t}\mathcal{U}] = \Re e[\sqrt{2}e^{j\omega t}R\mathcal{I}]$$

$$\Downarrow$$

$$\Re e[\sqrt{2}e^{j\omega t}\left\{\mathcal{U} - R\cdot\mathcal{I}\right\}] = 0 \qquad \Rightarrow \qquad \boxed{\mathcal{U} = R\cdot\mathcal{I}}$$